sustainability

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The Effect of Environmental Protection-Related Media Coverage on Corporate Green Innovation

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Abstract: Climate change and environmental pollution pose urgent global challenges that critically affect economic and social development. Corporate green innovation has emerged as a pivotal driver of coordinated environmental and economic progress. However, the existing research has predominantly focused on the effects of media coverage pertaining to corporations themselves in regard to corporate decisions, providing limited insight into the effects of media coverage on environmental protection. Thus, this study aims to investigate the governance effect of the media and its influence on corporate green innovation. Utilizing fixed-effect models, which involves analyzing data from various secondary sources, this study focused on Chinese manufacturing companies listed on the Shanghai and Shenzhen stock exchanges from 2008 to 2019. The results revealed that media coverage of environmental protection had a positive impact on corporate green innovation. In addition, this study found that industry turbulence enhanced the connection between media coverage of environmental protection and green innovation, whereas factor market development and managers' overseas experience weakened this relationship. Recognizing the role that the media plays in promoting green innovation can empower companies to effectively address environmental challenges and contribute meaningfully to sustainable development.

Keywords: media coverage of environmental protection; green innovation; industry turbulence; factor market development; manager's overseas background

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Citation: Cai, C.; Fan, M.; Dong, X.; Zhang, J. The Effect of Environmental Protection-Related Media Coverage on Corporate Green Innovation. *Sustainability* **2024**, *16*, 10887. https://doi.org/10.3390/su162410887

Academic Editor: Mário José Baptista Franco

Received: 27 September 2024
Revised: 13 November 2024
Accepted: 10 December 2024
Published: 12 December 2024

1. Introduction

Climate change poses a critical challenge to the world, with global warming leading to frequent extreme weather events and subsequent ecological and environmental problems. These issues have had serious impacts on global economic and social development. Corporations, particularly those in the energy and manufacturing sectors, are recognized as crucial players in addressing environmental concerns, and their activities are considered the main driving force behind environmental issues [1,2]. Corporations are increasingly acknowledging the substantial impact of environmental issues on their operations and are strategically positioning themselves with regard to environmental considerations [3]. Furthermore, owing to escalating environmental risks, corporations can strengthen their environmental responsibilities by adopting eco-friendly processes [4]. Thus, environmental sustainability has become a strategic priority for most corporations, rather than an isolated or optional agenda item.

Amid the acceleration of innovation [5], green innovation has become a propelling force behind aligned environmental and economic progress, critically supporting the reduction of environmental pollution [6]. Green innovation provides eco-friendly production technologies, facilitates the transformation and upgrading of industrial structures, promotes sustainable production methods, reduces natural resource consumption per unit of output, enhances energy utilization efficiency, and enables substantial savings and resource recycling [7]. Due to growing concerns regarding environmental issues, more companies are engaging in environmental innovation practices and are actively pursuing green innovation. Existing research has extensively explored various factors that influence green innovation, including institutional factors, corporate characteristics, the role of different stakeholders, and other types of factors [8,9].

In addition to these factors, mass media has gained widespread attention from scholars because it is an effective medium for information dissemination, substantially influencing the general public, companies, and policymakers. Media coverage acts as a social norm that shapes corporate behavior, whereas news media acts as a conduit for institutional pressure, urging companies to conform to the prevailing institutional logic [10]. When encountering institutional pressure, companies may seek to maintain legitimacy by actively engaging in environmental activities, focusing on adopting socially valuable practices within an institutional field, such as promoting green innovation. Media governance encompasses both supervisory and information dissemination effects [11,12]. Existing research has primarily explored the influence of media coverage on the implementation of corporate green behaviors; however, such coverage or attention often revolves around the companies themselves, including information disclosure by publicly listed companies, media exposure of CEOs, and reporting on corporate pollution events [13,14] (see Table 1). In contrast, the impact of media coverage of environmental protection on corporate behavior, such as green innovation, remains insufficiently explored.

Table 1. The literature on the role of the media and its impact on green behaviors.

References	IV	DV	Key Findings
Gao and Tran [15]	Media attention	Environmental performance	The higher the amount of media attention, the stronger the regulatory efforts by the company, leading to better environmental performance.
Kong, Kong, and Wang [13]	Media attention	Firms' environmental protection efforts	Media attention prompts companies to put more effort into environmental protection and elicits a more proactive response when facing local media attention and negative media coverage.
Wang, et al. [16]	CEO media exposure	Green technological innovation	CEO media exposure enhances the decision-making process in terms of green technological innovation in Chinese pollution-intensive companies.
Zhang, et al. [17]	Media coverage	Green technological innovation	The focus of the media solely exerts a markedly beneficial influence on the integration of green technological innovation, and it affects the output of green technological innovation only when combined with a certain level of marketization.
Wang and Zhang [14]	Media coverage on pollution incidents	Subsequent environmental investment	Media reporting on corporate pollution incidents has a significant positive influence on the likelihood of subsequent green acquisitions by companies.
Chen, et al. [18]	Media coverage	Green innovation	A favorable media depiction serves a more pronounced function in propelling corporate green innovation forward.
He, et al. [19]	Media attention	Green technological innovation	Media coverage, whether positive or negative, as well as environmental information disclosure, substantially influence corporate green advancements. Adverse media coverage has greater power to influence company behavior than positive coverage.

Table 1. *Cont.*

References	IV	DV	Key Findings
Hu and Wu [20]	Media pressure	Green technological innovation	Media pressure effectively enhances a company's green technological innovation.
Li, et al. [21]	New media attention	Green technological innovation	The new media environment incentivizes highly polluting companies to meet stakeholders' demands, leading to significant improvements in regard to their green technological innovation.

To address this research gap, our study aimed to answer the following two research questions and fill the void by examining the governance effect of the media and employing institutional theory: (1) How does media coverage of environmental protection information affect corporate green innovation? (2) What boundary conditions influence the relationship between media coverage of environmental protection and green innovation? To answer these questions, we analyzed data from multiple secondary sources. The findings indicate that media coverage of environmental protection positively influences innovation in regard to environmentally sustainable corporate practices. In addition, industry turbulence enhances the connection between media coverage of environmental protection and green innovation, whereas factor market development and managers' overseas backgrounds weaken this relationship.

The contribution of this study to the literature is threefold. First, concerning corporate green innovation, it expands upon prior research on the antecedents of green innovation by incorporating the mechanism of media governance effects, thus providing supplementary insights. Unlike prior studies that have mainly focused on the media coverage of companies, our research directs its focus toward examining the impact of media coverage on the sustainability of the external environment and corporate green behavior. This shift underscores the pivotal role of media coverage in regard to the dissemination of green information. Second, from the perspective of media coverage of environmental protection, our study enriches the relevant literature on the outcome variables of media coverage of environmental protection by providing additional evidence on the governance effect of the media. Prior research has primarily examined the impact of media coverage of climate change on government policymaking [22,23], with limited research linking media coverage of environmental protection to corporate behaviors. Hence, our study addresses this gap and explores the potential relationship between media coverage and corporate behavior. Finally, our study examines the boundary conditions influencing the connection between media coverage and corporate green behavior. Media coverage can address the problem of information asymmetry in turbulent environments, whereas the level of factor market development and managers' overseas backgrounds have substitution effects on the impact of the information being relayed in media coverage of environmental protection.

2. Conceptual Background

2.1. Motivation for Green Innovation

Green innovation occurs as a result of the combination of innovation and environmental protection efforts. It refers to the various technologies, processes, and products used to improve environmental quality and achieve renewable energy and resource utilization. It combines environmental protection with economic development to address environmental pollution and achieve sustainable development [24]. Chen, et al. [25] divide green innovation into product and process innovations, including hardware and software innovations for energy conservation, environmental protection, recycling, and green product design.

In contrast to traditional innovation, green innovation has positive externalities in terms of the knowledge and technological spillover effects. Specifically, leading companies engage in green technology R&D, improvements, and market production, benefiting from their early-bird advantage to capture niche markets, thereby motivating competitors to imitate or reverse the R&D taking place and, ultimately, improving the level of R&D in the industry. This process is based on leading companies alone bearing the cost of technology R&D, which generates technology spillover effects within the industry, resulting in positive externalities [26].

Based on a literature review, this study identified four main motivations for companies to greenify: competitive, legitimation, ecological responsibility, and managerial motivation. Competitive motivation refers to the belief that greenification enhances long-term profitability, that is, greenification can create competitive advantages and improve a company's financial performance [27]. Legitimation motivation refers to the belief that adopting sustainability measures can enhance the legitimacy of an organization, whereas proactive environmentalism enables companies to gain the necessary resources, reduce uncertainty, and ultimately enhances their survival capabilities. This motivation focuses on compliance with regulations and institutional norms [27–29]. The ecological responsibility motivation stems from an ethical and moral standpoint and the organization's view that they have a moral obligation to society, which assumes that greenification is the right thing to do [28]. Managerial motivation is an internal driving factor. Dynamic interactions exist between internal and external driving factors and behaviors [30]. External institutional and stakeholder pressures motivate managers to implement greenification behaviors to ensure the sustainable development of the company [31].

2.2. Drivers of Green Innovation

Green innovation has substantial strategic value for the sustainable development of both enterprises and society [32]. Given their significance, scholars have investigated the actors impacting green innovation within business enterprises. Studies have explored institutional-level factors, showcasing how environmental regulations, pressure from non-governmental organizations, government support, and intergovernmental political competition affect green innovation [33,34]. Scholars have demonstrated the impact of corporate governance, organizational resources, and financing constraints on green innovation, considering the characteristics of firms [35–37]. At the stakeholder level, research has indicated that pressure from different stakeholders, such as consumers, suppliers, and competitors, is a decisive factor in corporate green innovation [9,38,39]. At the individual level, research has emphasized the characteristics of managers, such as the CEO's foreign experience [40], CEO hubris [41], and CEO media exposure [16]. In addition, the literature has explored how media supervision affects corporate green innovation. Studies have found that media supervision makes the environmental information on companies transparent. Under the pressure of transparency brought by media coverage, companies enhance their environmental governance level [42]. Media supervision can also indirectly influence the innovation capabilities of corporate green technology, by impacting the behavior of external stakeholders [43]. The greater the level of media coverage, the higher the environmental pressure on companies, which, in turn, leads to increased levels of green innovation [16].

Although previous research has explored the influence of media reporting on corporate green innovation, these reports have mainly focused on the companies themselves, including the disclosure of information on listed companies, CEO media exposure, and corporate pollution incidents [13,16,18]. However, research on how media reporting on the environment affects corporate behavior is limited [44]. Therefore, this study mainly examined how environmental media reporting affects corporate green innovation.

2.3. The Role of Mass Media

As a carrier tool for information in the information age, mass media collects and processes information for the public, provides rapid information-based feedback, reduces the cost of information collection, and enables the public to conveniently access diverse information. The mechanism of the media governance effect primarily manifests in regard to two aspects. Media attention plays a supervisory role, whereas media coverage supervises and restrains the behavior of relevant corporate stakeholders, thereby improving corporate governance [11]. Furthermore, the media plays a role in information dissemination. As an information intermediary, the media can expand the scope of information dissemination, increase the speed of information dissemination, reduce the search cost for information seekers, and improve the level of information in the market [12]. The media primarily plays a supervisory role through information dissemination. Through information intermediation, the media expands the amount of available information, thereby influencing the audience's understanding of the information or changing the conditions of information dissemination [45]. Due to the role of the media as an external supervisor, the public makes corresponding reactions to the market based on the disseminated news reports, thereby influencing the daily governance behavior of corporations [46].

Based on the growing attention to environmental issues, the media have begun to extensively and increasingly disseminate relevant information about sustainable development [15,47–49]. The media has been increasingly recognized for its critical influence on public perceptions, which involves determining the quantity and nature of sustainable development information received by the public [50]. Furthermore, public perceptions shaped by the media critically affect the formulation of sustainable development-related government decisions and policy agendas [22,23]. Previous research has linked media coverage of sustainable development with the government as a subject. However, limited research has explored the link between media coverage of sustainable development and corporations, especially considering the long-term performance of corporations. Therefore, this study explored this issue and attempted to uncover the potential relationship between the two.

3. Hypotheses Development

3.1. Media Coverage of Environmental Protection and Green Innovation

As external stakeholders, media reporting can function as a social norm, exerting a certain level of influence on corporate behavior. Institutional theory suggests that corporate behavior is often driven by institutional pressure aimed at constructing and maintaining legitimacy [51]. News media, serving as a conduit for such pressure, can encourage corporations to adhere to the prevailing institutional logic [10]. When encountering institutional pressure, corporations may actively engage in environmental activities to maintain their legitimacy, leading them to adopt socially valuable practices within their institutional fields.

Institutional theory asserts that organizations are embedded in an institutional environment that encompasses symbolic and behavioral systems consisting of regulative mechanisms, constitutive rules, and normative rules that define shared meaning systems and generate distinct behavioral and action patterns [52]. Governments, professional associations, public opinion, and the media are prominent representatives of the institutional environment [53]. According to this theory, the institutional environment establishes implicit and explicit norms in regard to areas such as organizational structure and behavior that are expected to be met by members of an institutional field. Norms exert influence by setting standards that ensure appropriate behavior among members of a group [54]. Corporations conform to the prevailing norms of their institutional environment to reduce uncertainty and acquire the resources necessary for survival and success.

Institutional theory primarily focuses on understanding how social conformity shapes organizational behavior [55]. Organizations are viewed as seeking approval and are, thus, susceptible to social influence. Conformity encompasses not only meeting formal legal obligations, but also conforming to informal social norms. Hence, corporations respond to expectations beyond formal standards and legal regulations [56]. Formal mandatory regulations usually establish minimal environmental protection requirements and corporations may adopt environmental practices aimed at meeting formal environmental regulations. Furthermore, informal standards may lead corporations to comprehensively adopt green innovations.

In summary, media coverage of environmental protection generates awareness and societal norms that discourage sacrificing the environment for the benefit of an entity. In a media-driven institutional environment, corporations are influenced by institutional pressure to conform socially, maintain legitimacy, and reduce uncertainty to ensure their survival and growth. Therefore, corporations may respond positively to media coverage of environmental protection by actively embracing environmental innovations, specifically green innovations, as explored in this study. Building on this, we formulated the following hypothesis:

H1. *Media coverage of environmental protection positively influences corporate green innovation.*

Based on the main findings, and in line with previous research, this study introduced three potential moderating variables that could influence the connection between media coverage of environmental protection and corporate green innovation. Industry turbulence plays a crucial role in regard to the consideration of external factors, from a firm's perspective. Companies operating in turbulent industries must intensify their innovation activities and implement innovation strategies appropriately to remain competitive [57]. In addition, the effectiveness of corporate innovation can be enhanced in such environments [58]. Furthermore, when market mechanisms are fully effective, a firm's innovation capabilities gradually become apparent [59], and if resources are directed toward emission reduction and energy conservation, the resulting technological advancements can foster green innovation [60]. Shifting the focus to internal factors, from a firm's perspective, the widespread institutionalization of corporate social responsibility, environmental protection, and green innovation concepts in developed regions have contributed to the emergence of norms surrounding social responsibility and environmental protection. Consequently, managers with experience in these regions are more inclined to acknowledge the importance of social responsibility and environmental protection, thus facilitating the promotion of green innovation [40,61]. In summary, this study posits that industry turbulence, factor market development, and overseas managers' backgrounds, moderate the effects of media coverage of environmental protection on corporate green innovation. Subsequent subsections elaborate on the mechanisms underlying the role of each moderator in detail.

3.2. Industry Turbulence

Industry turbulence indicates shifts in customer preferences for products and services within an industry and is the underlying cause of environmental turbulence [62]. These changes are primarily driven by the volatility and uncertainty of the external environment [63]. Environmental turbulence within an industry pertains to the extent of the shifts in the market and/or technological landscape within a particular sector. According to contingency theory, firms are encouraged to adjust their strategies to align with changing external conditions, enabling them to effectively seize market opportunities [64]. Firms must initially gain a thorough understanding of their customers to achieve a competitive advantage by harnessing customer value. This understanding is developed by monitoring and analyzing the industry in which they operate [65]. In such cases, firm innovation becomes crucial for meeting ever-evolving customer demands when an industry experiences significant turbulence [66]. Firm innovation can be even more effective when market turbulence is particularly pronounced [58].

Environmental turbulence is characterized by rapid fluctuations in the linkages between specific actions and crucial performance indicators [67]. For instance, in environments where critical unforeseen events emerge swiftly, the data available to strategic decision-makers are often incomplete or outdated [67,68]. Decision-makers in turbulent environments frequently encounter challenges in identifying robust causal relationships because key decision variables may be unreliable or unavailable [69]. Given the difficulties faced by strategic decision-makers in turbulent industry environments in regard to collecting and processing the necessary information to evaluate causal relationships, it becomes challenging to formulate decision-making strategies that positively impact the firm's performance. In such circumstances, the role of the media in terms of information dissemination has become increasingly prominent. Industry turbulence reinforces the favorable effects of media coverage of environmental protection on firms' green innovation. Timely and accurate media reporting and media coverage of environmental protection reflects changes in the external environment, effectively reducing the information asymmetry resulting from industrial environmental turbulence. This enables firms to access more consistent and stable information, empowering decision-makers to process information rapidly and make informed decisions, thereby promoting firms' green innovation. Hence, we proposed the following hypothesis:

H2. *Industry turbulence positively moderates the relationship between media coverage of environmental protection and a firm's green innovation.*

3.3. Factor Market Development

Factor market development encompass the availability of financial resources, skilled labor, advanced technology, and other factor market resources allocated through market-regulated pricing [70]. As factor markets improve, the market becomes the primary driver of resource allocation [71]. Enterprises in regions with underdeveloped factor markets prioritize resource acquisition. However, as the market matures, all enterprises within the region can access factor resources, thus reducing the emphasis on resource acquisition [72]. Developed factor markets also establish fair environments and minimize biased resource allocation [73]. A sound and mature factor market system enhances resource agglomeration effects, facilitating the flow of capital, labor, and technology factors to enterprises, with higher efficiency in regard to factor allocation. A more developed regional capital market can provide funding support for innovative activities and alleviate insufficient R&D funding for enterprises. Likewise, a well-functioning labor market facilitates the mobility of highly knowledgeable workers across industries and enterprises, promoting knowledge and technology spillovers and enhancing enterprises' innovation output. A mature and active technology transfer market accelerates the diffusion of advanced technology, reduces the time required for innovation, and increases the speed and quantity of innovation outputs.

When market mechanisms are fully utilized, enterprises' innovation capacity gradually emerges [59]. Allocating resources to energy conservation and emission reduction can yield several positive effects, including driving green innovation, adopting eco-friendly production methods, and reducing pollutant emissions. In turn, this helps mitigate environmental pollution [60]. Therefore, enterprises in regions with advanced factor market development have more resources for green innovation. However, enterprises may proactively invest in innovation prior to changes in the external environment to gain a first-mover advantage. This weakens the informational impact of media coverage on the environment, indicating that the development of factor markets has a substitutive effect on the media's sustainability reporting. Therefore, we proposed the following hypothesis:

H3. *Factor market development exerts a dampening effect on the association between media coverage of environmental protection and a firm's green innovation.*

3.4. Manager's Overseas Background

Managers' values and cognition are substantially affected by their overseas background. Research has demonstrated that returning managers influenced by their experiences in developed countries and regions tend to exhibit higher levels of ethics and social responsibility. For instance, returnee managers and directors often guide their companies to increase investments in corporate social responsibility activities [9,74]. As concepts such as responsibility and environmental protection are firmly ingrained in developed countries and regions, managers with experience in these regions are more inclined to regard these concepts as established norms [40]. They internalize a strong commitment to environmental ethics and strive to enhance their company's environmental performance, thereby promoting green innovation [61]. Furthermore, prior studies have indicated that overseas experience enhances CEOs' overall capabilities and contributes to the development of green innovation. Managers' international experience enriches corporate governance and management practices by broadening their professional knowledge globally [40]. It also enhances their cognitive abilities, including ingenuity, solution finding, leadership, and information processing [75]. Hence, managers with overseas backgrounds are more inclined to drive corporate green innovation.

Considering the four primary motives for "greening" mentioned earlier, managers with overseas backgrounds in developed countries/regions are likely to possess strong managerial motivation to promote green innovation within their companies. Compared to external stimuli, these managerial motivations, as internal driving factors, enable companies to adopt a proactive and efficient approach to environmental innovation practices. In this context, it is anticipated that the influence of media coverage of environmental protection on corporate green innovation is diminished, indicating a substitution effect of managers' overseas backgrounds on the media's environmental reporting. Therefore, we proposed the following hypothesis:

H4. *The overseas background of managers negatively moderates the relationship between media coverage of environmental protection and a firm's green innovation.*

In summary, our conceptual model is shown in Figure 1.

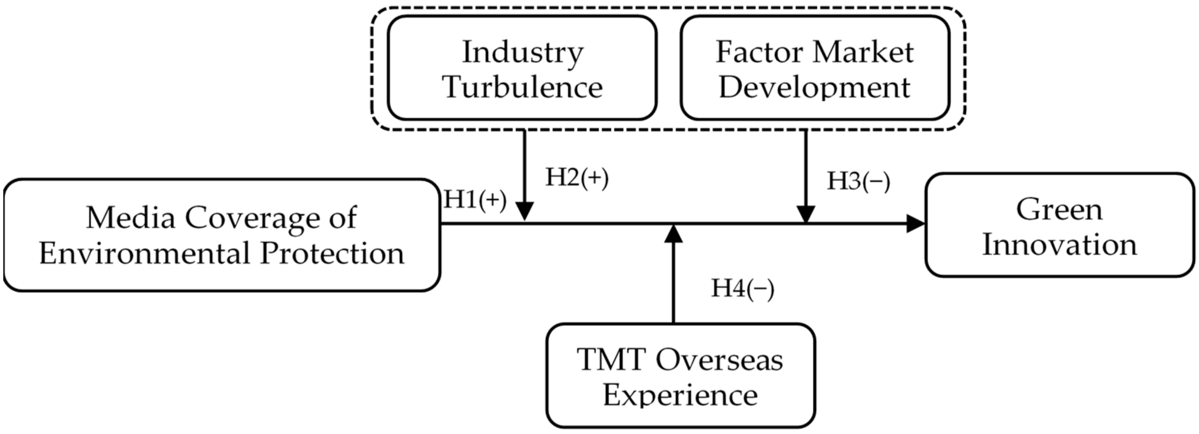

Figure 1. Conceptual model.

4. Methodology

4.1. Data and Sample

Our preliminary sample included all Chinese manufacturing companies listed on the Shanghai and Shenzhen stock exchanges between 2008 and 2019. We chose 2008 as our starting year because the data on one of our moderators, namely factor market development, are available for that year. We selected 2019 as our endpoint year because of the substantial disruption caused by the COVID-19 pandemic to most businesses in 2020. Furthermore, we excluded some samples according to the following four steps. First, firm-year observations under special treatment were excluded as their financial variables differed considerably from those of the other firms. Second, firm-year observations with missing data were excluded from the analysis. Third, firms with only one year of observation were excluded. Finally, to mitigate potential reverse causality, we lagged all explanatory variables by one year. Our independent variable ranged from 2008 to 2018 and the dependent variable covered the period from 2009 to 2019. Our final sample comprised 13,538 firm-year observations from 2010 firms. All the continuous variables were Winsorized at the 1st and 99th percentiles to mitigate the impact of extreme values.

Our data were obtained from multiple databases. We obtained data on our main dependent variable from the Green Patent Research Database (GPRD), a sub-database as part of the Chinese Research Data Services Platform (CNRDS), which is a widely used Chinese academic research database. The R&D intensity data were sourced from WIND, another well-known financial database in China. The data for all the other variables, including the independent variables, were obtained from the China Stock Market and Accounting Research (CSMAR) database. All these databases have been commonly utilized in previous research.

4.2. Measurements

Dependent variable: *Green Innovation.* Green innovation can be measured in various ways, including through the use of questionnaires, content analysis of annual and social responsibility reports, environmental research and development, and green patent applications [44,76,77]. Green patent applications are recognized as crucial indicators of green innovation because of their robustness [76]. Therefore, we chose the number of green patent applications as our green innovation indicator. To address data skewness, we calculated green innovation by taking the natural logarithm of the number of green patent applications plus one. In addition, we employed the number of green patent applications for the utility models and the green invention patent applications as proxies of green innovation in the robustness check.

Independent variable: *Media Coverage of Environmental Protection (MCEP).* Media coverage of environmental protection was measured as the natural logarithm of the number of news articles whose titles were related to green, ecology, or environmental protection. We collected all the news titles from the CNRDS database between 2008 and 2018. Drawing on the approach by Li, et al. [78], we considered the news titles containing keywords such as "environmental protection," "low carbon", "low energy", "emissions reduction", "energy saving", "clean energy", "circular economy", "cyclic utilization", "ecological balance", "afforestation project", "windbreak forest", and "sustainable use", as environmental protection-related news. To identify media coverage of environmental protection, we searched for these related keywords in news titles, denoting a news article as environmental protection-related if one of the keywords was present (coded as 1), otherwise it was marked 0. Finally, we tallied the number of environmental protection-related articles in a year as the amount of news related to green, ecology, or environmental protection. We considered the number of news articles whose content was related to green, ecology, or environmental protection as an alternative measurement.

Moderators: *Industry turbulence.* Industry turbulence reflects the extent to which customer demand and preferences change. Following the work by Shen, Zhou, Wang, and Zhang [70], we first conducted a regression of industry sales revenue for the current year and the previous four years, with a time variable ranging from 1 to 5. Next, we measured industry turbulence as the ratio of the standard error of the regression coefficient to the average industry sales for those years.

Factor market development. We used one dimension of the marketization index developed by the NERI to measure factor market development [79,80]. Factor market development encompasses three dimensions: the marketization of the financial sector (including competition in the financial sector and marketization of credit fund distribution), the availability of labor resources (resident/household population), and the marketization of technological achievements. A higher index value indicates a greater degree of marketization in the factor market. The data on factor market development were only available from 2008 to 2016. We extrapolated the data to 2018, based on the average growth from 2008 to 2016.

Top management team (TMT) overseas experience. TMT overseas experience was coded as a binary variable, with a value of 1 if any of the current directors, supervisors, or senior managers had experience of overseas study or employment, otherwise they were marked as 0.

Control variables: A set of variables that may affect a firm's green innovation was controlled for. First, we controlled for two basic variables related to listed companies that might have considerably affected a firm's strategies: the firm's size and the firm's age. The *firm size* was quantified as the natural logarithm of the employee count. The *firm age* was calculated as the natural logarithm of the number of years since a firm's listing. Second, we controlled for three variables related to a firm's financial situation: the return on equity (ROE), firm leverage, and firm growth. The *ROE* was determined by dividing the net profit by the equity. The *firm leverage* was calculated as the ratio of long-term debt to the total assets. The *firm growth* was calculated as the ratio of current-year sales revenue to the previous year's sales revenue minus one. Third, because our main dependent variable was green innovation, which is highly related to R&D investment, we controlled for *R&D intensity,* measured by dividing the R&D investment by the sales revenue in the same year. Fourth, we controlled for one variable related to corporate governance. *The largest shareholder* was measured using the shares held by the largest shareholder. Fifth, we controlled for one board-level variable. The *board size* was also included as a control variable, represented by the natural logarithm of the number of board members. Finally, we controlled for a regional level variable, namely *factor market development*, which was measured by one dimension of the marketization index created by the NERI [79]. Detailed measurements of all variables are shown in Table 2.

Table 2. Variable measurements.

Variables	Measurement	Data Source
Green Innovation	The natural logarithm of the number of green patent applications plus one.	CNRDS
Media Coverage of Environmental Protection (MCEP)	The natural logarithm of the number of news items of which the title is related to green, ecology, or environmental protection. An item of news is considered as environmental protection-related news if the title contains the keywords "environmental protection", "low carbon", "low energy", "emissions reduction", "energy saving", "clean energy", "circular economy", "cyclic utilization", "ecological balance", "afforestation project", "windbreak forest", or "sustainable use".	CSMAR
Industry Turbulence	The ratio of the standard error of the regression coefficient to the average industry sales for that year. Here, the regression is $Y_t = \beta_0 + \beta_1 t$, where $t = 1, 2, 3, 4, 5$, Y_t is the industry sales for the present year and prior four years.	CSMAR
Factor Market Development	Factor market development is one dimension of the marketization index developed by the NERI.	NERI
TMT Overseas Experience	It is coded as 1 if any of the current directors, supervisors, or senior managers have an overseas background, including past and present overseas study or employment, otherwise it is coded as 0.	CSMAR
Firm Size	The natural logarithm of the number of employees.	CSMAR
Firm Age	The natural logarithm of the years since the firm was listed plus one.	CSMAR
ROE	The proportion of the net profit to the equity.	CSMAR
Firm Leverage	The rate of long-term debts to total assets.	CSMAR
Firm Growth	The ratio of the sales revenue in the current year to that in the previous year minus one.	CSMAR
R&D Intensity	The ratio of R&D expenditure to the sales revenue.	WIND
Largest Shareholder	The number of shares held by the largest shareholder.	CSMAR
Board Size	The natural logarithm of the number of board members	CSMAR

4.3. Model Specification

We conducted a series of two-way fixed-effect models using industry dummies to examine our hypotheses. The model specifications are as follows:

$$\begin{aligned} Green\ Innovation_{it+1} \\ &= \beta_0 + \beta_1 MCEP_{it} + \beta_2 Industry\ Turbulence_{it} \\ &+\beta_3 Factor\ Market\ Development_{it} + \beta_4 TMT\ Oversea\ Experience_{it} \\ &+\beta_5 MCEP_{it} \times Industry\ Turbulence_{it} \\ &+\beta_6 MCEP_{it} \times Factor\ Market\ Development_{it} \\ &+\beta_7 MCEP_{it} \times TMT\ Oversea\ Experience_{it} + CV + \sum Year \\ &+\sum Firm + \sum Industry + \varepsilon_{it} \end{aligned}$$

where CV denotes the control variables, including the *Firm Size*, *Firm Age*, *Roe*, *Firm Leverage*, *Firm Growth*, *R&D Intensity*, *Largest Shareholder*, and *Board Size*. *Year*, *Firm*, and *Industry* denote the year, firm, and industry fixed effects.

To reduce multicollinearity in the model, we mean centered the independent variables and moderators before generating the interaction terms.

5. Results

5.1. Hypotheses Testing

Table 3 presents the descriptive statistics and correlations of all the variables considered in this study. The correlation coefficients presented in Table 3 indicate a positive relationship between media coverage of environmental protection and green innovation (r = 0.03, $p < 0.05$).

Table 3. Descriptive statistics and correlations (N = 13,538).

No.	Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
(1)	Green Innovation(t + 1)	1												
(2)	MCEP(Title)	0.03 *	1											
(3)	Industry Turbulence	−0.05 *	0.06 *	1										
(4)	Factor Market Development	0.06 *	0.04 *	−0.16 *	1									
(5)	TMT Overseas Experience	0.06 *	−0.01	−0.05 *	0.14 *	1								
(6)	Firm Size(log)	0.28 *	0.01	−0.04 *	−0.04 *	0.05 *	1							
(7)	Firm Age(log)	0.03 *	0.02 *	−0.11 *	0.04 *	−0.06 *	0.41 *	1						
(8)	ROE	0.10 *	0.03 *	0.02 *	0.00	0.06 *	0.11 *	−0.12 *	1					
(9)	Firm Leverage	0.16 *	0.01	0.02 *	−0.08 *	−0.05 *	0.46 *	0.40 *	−0.20 *	1				
(10)	Firm Growth	0.03 *	0.07 *	−0.01	0.02 *	0.04 *	0.03 *	−0.07 *	0.30 *	0.02 *	1			
(11)	R&D Intensity	0.13 *	0.02 *	−0.15 *	0.22 *	0.10 *	−0.24 *	−0.23 *	−0.04 *	−0.30 *	−0.02 *	1		
(12)	Largest Shareholder	0.02 *	−0.03 *	0.06 *	−0.03 *	0.00	0.16 *	−0.12 *	0.12 *	0.03 *	0.01	−0.10 *	1	
(13)	Board Size(log)	0.09 *	−0.01	0.05 *	−0.10 *	0.05 *	0.26 *	0.15 *	0.04 *	0.17 *	−0.01	−0.14 *	−0.01	1
	Mean	0.56	7.79	0.03	6.75	0.55	7.74	2.02	0.07	0.40	0.18	0.04	0.34	2.13
	S.D.	0.94	0.56	0.02	2.23	0.50	1.11	0.78	0.11	0.19	0.35	0.03	0.14	0.19

Note: * $p < 0.05$.

A fixed-effect model was used to further investigate the influence of media coverage of environmental protection on a firm's green innovation. Model (1) encompassed all the control and independent variables. To assess the moderating effects of industry turbulence, factor market development, and TMT overseas experience on the link between media coverage of environmental protection and green innovation, a series of fixed-effect models were employed. First, Model (2) incorporated industry turbulence into Model (1). Subsequently, Model (3) included the interaction term between industry turbulence and media coverage of environmental protection. Next, Model (4) added factor market development to Model (1), followed by Model (5), which included the interaction term between factor market development and media coverage of environmental protection. Model (6) then incorporated TMT overseas experience into Model (1), whereas Model (7) included the interaction term between TMT overseas experience and media coverage of environmental protection. The regression results are shown in Table 4.

As illustrated by Model (1) in Table 4, the results indicate a notable and positive influence of media coverage of environmental protection on green innovation (b = 0.31, $p < 0.01$), strongly supporting H1. Furthermore, as shown in Model (3), the coefficient of the interaction term in relation to industry turbulence and media coverage of environmental protection is significantly positive (b = 1.29, $p < 0.05$), indicating that industry turbulence strengthens the positive relationship between media coverage of environmental protection and green innovation, thereby supporting H2. The simple slope analysis in the left panel of Figure 2 demonstrates a significantly positive relationship between media coverage of environmental protection and green innovation when industry turbulence is high (b = 0.34, $p < 0.01$). The effect is notably stronger than when industry turbulence is low (b = 0.29, $p < 0.01$).

Table 4. Regression results.

Variables	Dependent Variable: Green Innovation(t + 1)						
	Model (1)	**Model (2)**	**Model (3)**	**Model (4)**	**Model (5)**	**Model (6)**	**Model (7)**
Independent Variable							
MCEP(Title)	0.31 **	0.31 **	0.32 **	0.32 **	0.33 **	0.31 **	0.33 **
	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
Moderators							
Industry Turbulence		−0.08	−0.12				
		(0.40)	(0.40)				
Factor Market Development				−0.00	−0.00		
				(0.01)	(0.01)		
TMT Overseas Experience						0.01	0.02
						(0.02)	(0.02)
Interactions							
MCEP(Title) × Industry Turbulence			1.29 *				
			(0.61)				
MCEP(Title) × Factor Market Development					−0.01 *		
					(0.00)		
MCEP(Title) × TMT Overseas Experience							−0.04 +
							(0.02)
Control Variables							
Firm Size(log)	0.11 **	0.11 **	0.11 **	0.11 **	0.11 **	0.11 **	0.11 **
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
Firm Age(log)	−0.02	−0.02	−0.02	−0.02	−0.02	−0.02	−0.02
	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)	(0.03)
ROE	0.42 **	0.42 **	0.41 **	0.42 **	0.41 **	0.42 **	0.42 **
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Firm Leverage	0.19 **	0.19 **	0.19 **	0.19 **	0.19 **	0.19 **	0.19 **
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
Firm Growth	−0.03 +	−0.03 +	−0.04 *	−0.03 +	−0.04 *	−0.03 +	−0.03 +
	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)	(0.02)
R&D Intensity	0.85 *	0.85 *	0.85 *	0.85 *	0.85 *	0.84 *	0.84 *
	(0.41)	(0.41)	(0.41)	(0.41)	(0.41)	(0.41)	(0.41)
Largest Shareholder	0.10	0.10	0.11	0.10	0.10	0.10	0.10
	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)	(0.13)
Board Size(log)	0.06	0.06	0.06	0.06	0.06	0.06	0.06
	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)
Constant	−0.78 **	−0.78 **	−0.79 **	−0.79 **	−0.79 **	−0.78 **	−0.78 **
	(0.28)	(0.28)	(0.28)	(0.28)	(0.28)	(0.28)	(0.28)
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES
Observations	13,538	13,538	13,538	13,538	13,538	13,538	13,538
Number of Firms	2010	2010	2010	2010	2010	2010	2010
R-squared	0.05	0.05	0.05	0.05	0.05	0.05	0.05

Note: Robust standard errors in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Likewise, as evidenced in Model (5), the interaction term between factor market development and media coverage of environmental protection demonstrates a significant negative effect (b = −0.01, $p < 0.05$). This finding suggests that factor market development attenuates the positive relationship between media coverage of environmental protection and green innovation, thus supporting H3. The simple slope in the upper-right panel of Figure 2 shows that the relationship between media coverage of environmental protection and green innovation is significantly positive when factor market development is high (b = 0.30, $p < 0.01$), which is significantly lower than when factor market development is low (b = 0.35, $p < 0.01$).

Similarly, as evidenced in Model (7), the interaction term between TMT overseas experience and media coverage of environmental protection shows a significant negative impact (b = −0.04, $p < 0.10$). This indicates that the positive relationship between media coverage of environmental protection and green innovation is weakened by TMT overseas experience, thereby supporting H4. The simple slope analysis in the right panel of Figure 2 reveals that the connection between media coverage of environmental protection and green innovation is significantly positive when the TMT has overseas experience (b = 0.29, $p < 0.01$), which is notably lower than when the TMT has no overseas experience (b = 0.33, $p < 0.01$).

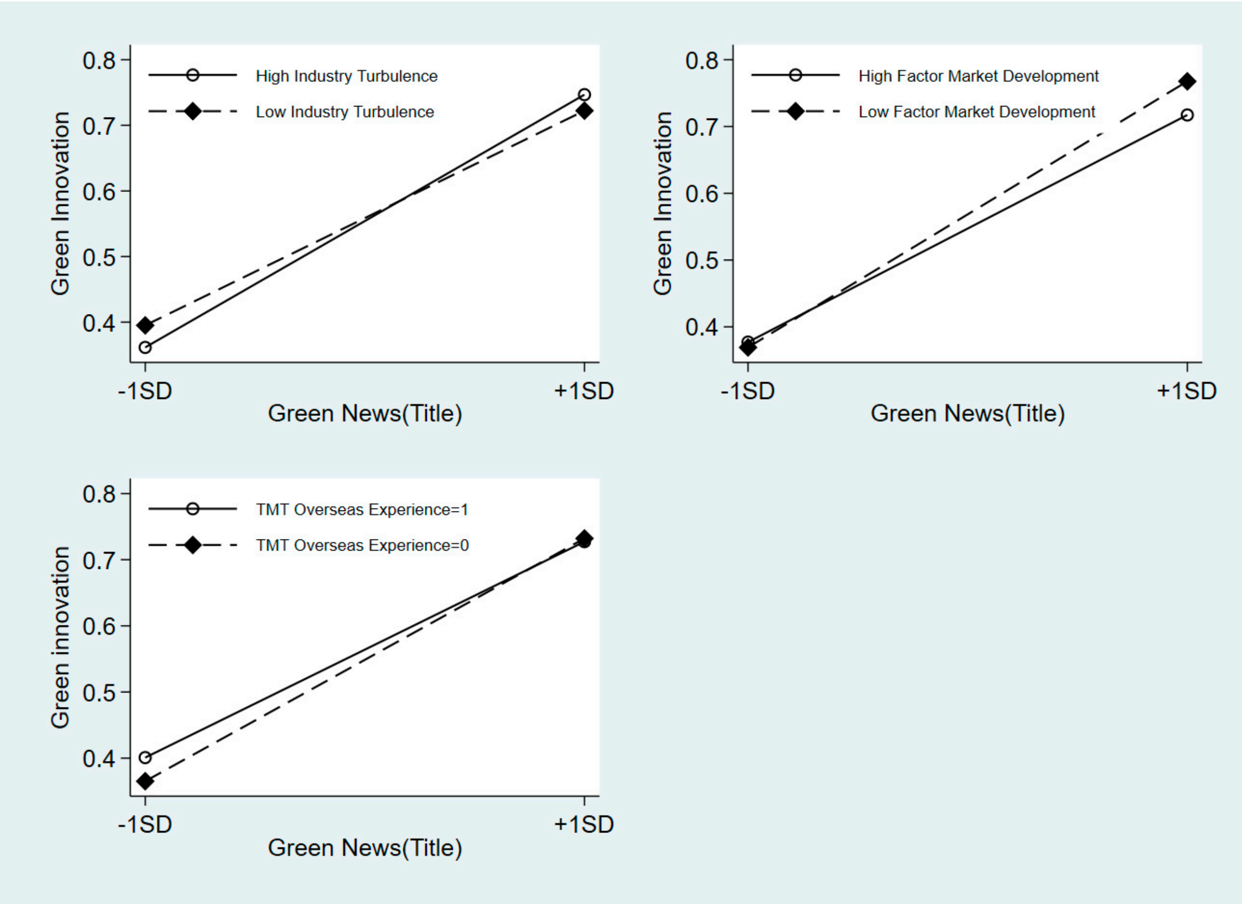

Figure 2. Simple slope analysis of the moderating effects.

5.2. Robustness Tests

Several robustness checks were conducted as follows.

An alternative measure of media coverage of environmental protection based on the news content was used. In our study, media coverage of environmental protection was quantified as the natural logarithm of the number of news articles with titles related to green, ecology, or the environment. However, the primary information conveyed by the news is through its content, and the public's reactions to the market are based on it. Therefore, we recalculated the number of environmental protection-related articles as the natural logarithm of the number of news articles whose content was related to green, ecology, or environmental protection. These results are consistent (see Table 5).

An alternative measure of green innovation using the applications for green invention patents was used. Green patents include both green invention and utility patents. Green invention patents involve novel technical solutions in regard to products, processes, and enhancements that focus on advancing environmental sustainability. As a different measure of green innovation, we examined the number of applications for green invention patents.

The conclusions derived from this alternative assessment align with our primary results (see Table 6).

Table 5. Robustness check (alternative measurement of media coverage of environmental protection).

Variables	Dependent Variable: Green Innovation(t + 1)						
	Model (1)	**Model (2)**	**Model (3)**	**Model (4)**	**Model (5)**	**Model (6)**	**Model (7)**
Independent Variable							
MCEP(Content)	0.79 **	0.79 **	0.80 **	0.82 **	0.84 **	0.78 **	0.81 **
	(0.15)	(0.15)	(0.15)	(0.16)	(0.17)	(0.15)	(0.15)
Moderators							
Industry Turbulence		−0.08	−0.21				
		(0.40)	(0.41)				
Factor Market Development				−0.00	−0.01		
				(0.01)	(0.01)		
TMT Overseas Experience						0.01	0.02
						(0.02)	(0.02)
Interactions							
MCEP(Content) × Industry Turbulence			1.53 +				
			(0.84)				
MCEP(Content) × Factor Market Development					−0.01 +		
					(0.01)		
MCEP(Content) × TMT Overseas Experience							−0.04
							(0.03)
Control Variables	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES
Observations	13,538	13,538	13,538	13,538	13,538	13,538	13,538
Number of Firms	2010	2010	2010	2010	2010	2010	2010
R-squared	0.05	0.05	0.05	0.05	0.05	0.05	0.05

Note: Robust standard errors in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

Table 6. Robustness check (alternative measurement of green innovation).

Variables	Dependent Variable: Green Innovation(t + 1)						
	Model (1)	**Model (2)**	**Model (3)**	**Model (4)**	**Model (5)**	**Model (6)**	**Model (7)**
Independent Variable							
MCEP(Title)	0.28 **	0.28 **	0.29 **	0.29 **	0.30 **	0.28 **	0.30 **
	(0.05)	(0.05)	(0.05)	(0.06)	(0.06)	(0.05)	(0.05)
Moderators							
Industry Turbulence		−0.03	−0.06				
		(0.33)	(0.33)				
Factor Market Development				−0.00	−0.00		
				(0.01)	(0.01)		
TMT Overseas Experience						0.01	0.01
						(0.02)	(0.02)
Interactions							
MCEP(Title) × Industry Turbulence			0.88 +				
			(0.51)				
MCEP(Title) × Factor Market Development					−0.01 **		
					(0.00)		
MCEP(Title) × TMT Overseas Experience							−0.03 *
							(0.02)
Control Variables	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES	YES	YES	YES
Industry FE	YES	YES	YES	YES	YES	YES	YES
Observations	13,538	13,538	13,538	13,538	13,538	13,538	13,538
Number of Firms	2010	2010	2010	2010	2010	2010	2010
R-squared	0.05	0.05	0.05	0.05	0.05	0.05	0.05

Note: Robust standard errors in parentheses; ** $p < 0.01$, * $p < 0.05$, + $p < 0.1$.

6. Discussion

This study investigated the connection between media coverage of environmental protection coverage and corporate green innovation, while also probing the boundary conditions that influence this relationship, based on the internal and external factors of the company. Specifically, previous studies have emphasized the role of the media in corporate governance, namely its supervisory role and in regard to information dissemination [10,11]. This study further extended this theory and the empirical analysis showed that media coverage of environmental protection has a positive impact on corporate green innovation (supporting H1). Secondly, industry turbulence has a positive moderating effect on corporate green innovation (supporting H2), which is consistent with Tsai, Yang, Tsai, and Yang [58]'s study, which found that in industries with significant market turbulence, corporate innovation is more effective. This study further suggested that in turbulent environments, timely media coverage can help corporate decision-makers respond quickly to external environmental changes, thereby promoting green innovation. Thirdly, factor market development has a negative moderating effect on green innovation in enterprises (supporting H3), meaning that in regions with developed factor markets, enterprises can more easily access innovation resources, reducing their dependence on media coverage information, which is consistent with Jia and Lin [59]'s study. Finally, the negative moderating effect of the top management team's overseas experience on corporate green innovation (supporting H4) is consistent with Quan, Ke, Qian and Zhang [40]'s study, which found that CEOs with foreign experiences tend to drive corporate green innovation. Our study further suggested that this internal motivation may weaken the impact of external media pressure on corporate green innovation.

The relevant findings and conclusions in this study provide meaningful theoretical and practical insights for both researchers and managers.

6.1. Theoretical Contributions

Our research contributes to theoretical development in regard to this topic and offers directions for future research. First, a growing body of research has investigated corporate green innovation and explored the influencing factors at various levels, including institutions, firm characteristics, and stakeholders [8,37]. Moreover, the impact of media attention on corporate green innovation has received considerable scholarly attention [20,45]. However, previous studies on the governance effects of the media have mostly focused on how media coverage of companies themselves influences their behavior, with little inquiry into how media coverage of environmental protection affects corporate conduct. Thus, this study adds value by looking into the impact of green media reporting on corporate green innovation, expanding the current literature on green innovation drivers and supplementing previous work.

Second, given the increasing media attention dedicated to environmental issues, which are inextricably linked to human health and social development, it is critical to investigate how firms respond to media signals about environmental change. While previous research has emphasized the impact of media-shaped public perceptions on government environmental policies [22], the relationship between green media reporting and firms' green actions has received little attention. This study addresses this gap by investigating the potential relationship between green media reporting and firms' green behavior, adding to the literature on green media reporting as an outcome variable and providing new insights into media governance impacts.

Finally, this study examined the conditional factors influencing the effect of media coverage of environmental protection on corporate green innovation: (1) Industry turbulence reflects the market or technological changes within the industry. In a turbulent environment, media reporting on environmental sustainability can assist decision-makers in developing a timely understanding of external environmental changes, thereby promoting firms' engagement in green innovation by addressing information asymmetry. (2) The degree of factor market development indicates the mobility and resource distribution capacity of production inputs. Areas with more advanced factor markets have more resources accessible for companies' green innovation, thereby diluting the media's informational influence. (3) Managers with overseas backgrounds typically exhibit a proactive approach to improving a firm's environmental ethics, thereby generating substitutive effects on green media reporting.

These theoretical contributions enhance our understanding of the interplay between media reporting, corporate green innovation, and the moderating factors that shape this relationship. They provide useful perspectives for guiding future research in the green innovation domain.

6.2. Managerial Implications

This research presents multiple salient implications for public media outlets and business leaders.

First, the media should effectively utilize its governance functions for monitoring and information distribution. As a vital institution in the media-driven institutional environment, it is crucial for the media to promptly, objectively, and efficiently disseminate information related to the environment, green initiatives, and ecology. By creating social norms for environmental protection and pollution reduction, the media can exert institutional pressure on enterprises to respond positively to green reporting and aggressively implement green innovation.

Second, enterprises should be alert to green information conveyed by the media. To achieve better development, enterprises must adhere to numerous institutional norms and embrace environmental innovation. This is crucial in situations typified by industry turbulence, where the external environment is volatile and unpredictable. Decision-makers frequently confront difficulties in gathering and digesting the essential information needed for strategic decisions that improve corporate performance. In regions with less developed factor markets, limited factor mobility and allocation efficiency prevent enterprises from obtaining sufficient resources, reducing their innovative capabilities. Furthermore, managers with little overseas experience in developed countries/regions may lack the general ability to improve corporate governance and management practices, as well as fail to comprehend the significance of environmental preservation for corporate development. As a result, enterprises may struggle to engage in green innovation in contexts typified by severe industry turbulence, low levels of factor market development, and managers without an overseas background. In such circumstances, corporate executives should pay close attention to media coverage, promptly respond to external environmental changes, and address the various challenges and obstacles encountered during the green innovation process.

6.3. Policy Implications

This study provides policy implications to assist the government in promoting green corporate innovation through increased media exposure.

First, the government should use various media channels to actively promote successful green innovation and related news cases. Broadcasting these positive and sustainable stories will encourage more firms to pursue sustainable efforts and eco-friendly technologies. For example, the government can intensively promote news on environmental protection-related policies and corporate green practices using self-media platforms. Simultaneously, by establishing close cooperative relationships with news outlets, the government can provide accurate information and expert opinions to help the media better understand the importance of media coverage of environmental protection. With a deeper understanding, the media can provide more in-depth coverage of relevant developments. Moreover, special funds can be set up to support newspapers, new media, and other media outlets to continuously track and report on corporate green innovation and allow more firms to enhance their views on the importance of green transformation.

Second, while leveraging the positive role of media coverage of environmental protection, the government should tailor policies based on the specific circumstances, considering industry turbulence, regional factor market development, and managerial backgrounds. Specifically, when the relevant companies are in relatively turbulent industries, have less developed factor markets, or have managers without overseas experience, the government can moderately increase the aforementioned policies on media coverage of environmental protection through the policymaking process, including self-media platform coverage and new media coverage, to consolidate the positive impact of media coverage of environmental protection on corporate green innovation. In contrast, if the companies are in relatively stable industries, have developed factor markets, or have internationally experienced management, the government can moderately reduce such policy guidance and allow more autonomy in regard to corporate green innovation. In summary, the government should adopt targeted policy tools based on different corporate realities to harness the role of media coverage of environmental protection in facilitating corporate green innovation.

6.4. Limitations and Future Research

First, this study took the media governance effect as a starting point, combined with institutional theory, to determine the mechanism of green media reporting, focusing on how green media reporting influences corporate green innovation. However, there were some limitations in regard to quantifying the independent and dependent variables in this study. Specifically, the independent variables in this study were only focused on mass media reporting. However, with the internet's swift evolution, social media has increased its influence on the public and information disseminated through social media has become an important aspect in regard to people's internet browsing, playing an increasingly crucial role in information dissemination. Future research should consider incorporating social media to explore its impact on corporate development. In addition, this study roughly measured the dependent variable, green innovation, using the number of green patents obtained by companies, without further differentiating between the various types. Future research could divide green innovation into input and output components or product and process innovation to explore the specific impacts of green media reporting on each segmented component in a more nuanced manner.

Second, this study focused on three boundary conditions in terms of the influence of green media reporting on green innovation. Future research could explore other boundary conditions, such as individual- and industry-level factors. At the individual level, political connections can enable companies to access low-cost resources [73] and such resource benefits may lead to path dependence for companies, hindering their ability to identify and integrate new operational technologies [70], which may further enhance the effect of green media reporting on green innovation. In addition, at the industry level, intense industry competition, characterized by numerous competing companies and fragmented market shares, intensifies companies' demands for factor resources. In this situation, companies may pay more attention to green media reporting, as it can promptly reflect external market changes and provide relevant information for companies to obtain the necessary resources.

Finally, the research sample in this study was limited to listed companies in China and the research context was confined to China. Therefore, the applicability of these findings to publicly listed firms in other contexts requires further investigation. In addition, this study did not explore the mediating mechanism of green media reporting on corporate green innovation, specifically, the variables through which green media reporting on green change are transmitted. Future research should investigate this issue in-depth.

Author Contributions: Conceptualization, C.C. and X.D.; formal analysis, M.F.; funding acquisition, C.C.; methodology, M.F.; resources, X.D.; software, C.C. and J.Z.; supervision, X.D.; visualization, J.Z.; writing—original draft, M.F., C.C., J.Z. and X.D.; writing—review and editing, C.C. and X.D. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the STU Scientific Research Institution Grant [grant number STF20013] and the Philosophy and Social Science Planning Program of Guangdong Province [grant number GD21YDXZGL01].

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data are available upon request from the authors.

Conflicts of Interest: The authors declare that there are no conflicts of interest.

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